EMOTION RECOGNITION MODEL USING FUSION IN MULTIMODAL CNN

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TABLE OF CONTENTS

1. ABSTRACT

Facial expression recognition is an intuitive task for humans but remains a complex challenge for computer algorithms due to the intricacies of human emotions and variations in facial features. Recent advancements in computer vision and machine learning, particularly convolutional neural networks (CNNs), have enabled significant progress in this domain. This paper proposes a novel technique called Facial Emotion Recognition using Convolutional Neural Networks (FERC), designed to detect emotions from images with enhanced precision and efficiency.

The FERC model consists of a two-part CNN architecture. The first part isolates the facial region by removing background noise, ensuring that only relevant information is processed. The second part focuses on extracting facial feature vectors, a crucial step for accurately identifying emotional cues. A key innovation of the FERC model is the use of the Expressional Vector (EV), which captures subtle nuances of facial expressions to classify emotions into five distinct categories.

By integrating these components, the proposed FERC model addresses common challenges such as background interference and feature generalization, delivering robust performance across diverse datasets. This research contributes to advancements in emotion recognition systems, paving the way for applications in fields such as healthcare, education, and human-computer interaction.

2. INTRODUCTION

Facial expressions are the key is to understanding the human behavior and how it reacts or interact with the environment. Computer interface provides technology which analyses human and computer interaction.

Emotion recognition systems use machine learning algorithms, sensors, and biometric analysis to decode human emotions from various cues such as facial expressions, voice intonations, and physiological signal.[2]

2.1 Data Collection

Sources: Data is gathered from specialized datasets such as FIR 2013, RAF-db,etc.

2.2 Recognition Models

Convolutional Neural Networks (CNNs): These are widely used for analyzing facial expressions in images and videos.

Recurrent Neural Networks (RNNs): Often used in combination with CNNs to capture temporal dynamics in video sequences.

Rule-based and Classical Machine Learning Approaches: Including Support Vector Machines (SVMs) and Naive Bayes classifiers are used for text-based emotion detection.

2.3 Techniques

Facial Expression Recognition: This technique uses computer vision to analyze facial expressions and identify emotions.

[3]Text-Based Emotion Detection: Natural Language Processing (NLP) techniques are used to analyze text data, such as social media posts or written communication, to identify emotional content. [3]

2.4 Challenges

Accuracy: Accuracy varies a lot in between different datasets.

[4]Viewpoint Variation: The image might look different if it's taken from an angle, which is called a variation in viewpoint.[4]

Inefficient Detections: During real-time testing, wearing glasses can cause issues for OpenCV eye detectors in accurately finding individual eyes.[4]

2.5 Applications

Educational sector, emotion recognition is being explored as a tool for personalized learning

Emotion recognition technology has become a pivotal tool in enhancing customer interactions across various sectors.

3. MOTIVATION

Facial emotions convey the intention of a person. The emotion communicates the state of the person such as joy, sadness, anger. The human communication has one-third part of verbal communication and two third part of nonverbal communication. Moreover, the facial expressions are an important means of interpersonal communication Thus, there exists a demand to develop a machine, which can recognize the human emotion and confidently convey said emotion to computers for

Enhanced Human-Computer Interaction: Emotion detection allows machines to understand and respond to human emotions, making interactions more natural and effective.

Applications in Various Fields: These models are used in healthcare, marketing, education, and public safety to improve services and outcomes.

Advancements in AI: Emotion detection drives progress in AI by integrating complex human emotional understanding into machine learning systems.

Data Utilization: With the rise of social media and digital communication, there’s a vast amount of data available for emotion analysis, providing valuable insights and improving user experiences.

4. LITERATURE SURVEY

Traditional Approaches to Emotion Recognition

4.1 Handcrafted Features and Machine Learning

Early methods relied on manual feature extraction from facial images, such as geometric features (e.g., distances between facial landmarks) or appearance features (e.g., texture, shape).

Algorithms such as Support Vector Machines (SVM), Random Forests, and k-Nearest Neighbors (k-NN) were popular choices for classification.

There were some limitations as these methods were sensitive to variations in lighting, pose, and occlusion, and required significant manual effort for feature engineering, resulting in low generalizability and poor performance in complex real-world scenarios.

4.2 Statistical Models

[9]**Hidden Markov Models (HMM)**: Used for modeling temporal dependencies in speech and video data. Effective in recognizing speech-based emotions but limited in facial emotion recognition due to complex facial dynamics.

[10]**Gaussian Mixture Models (GMM):** Applied for modeling facial expression distributions. They provided decent performance but were sensitive to data variations.

4.3 Emergence of Deep Learning Techniques

Convolutional Neural Networks (CNNs)

Architecture: CNNs automatically learn hierarchical features from raw pixel data, eliminating the need for manual feature extraction. They consist of convolutional, pooling, and fully connected layers.[1]

Key Contributions:

[5]**AlexNet:** Pioneered deep learning for visual tasks, inspiring the use of CNNs in facial emotion recognition.

[6]**VGGNet**: Enhanced feature extraction with deeper architectures and smaller convolutional filters, improving accuracy.

[7]**ResNet**: Introduced residual connections, enabling the training of much deeper networks, which improved performance on complex datasets like FER2013.

Limitations: Despite high accuracy on standard benchmarks, CNNs struggle with generalization in real-world scenarios with unseen data and varying conditions.

Recurrent Neural Networks (RNNs) and Long Short-Term Memory Networks (LSTMs)

Applications: RNNs and LSTMs are used to capture temporal dependencies in sequential data, such as video sequences or speech.

Advantages: They excel in recognizing dynamic emotions by considering the temporal evolution of facial expressions or vocal tones.

Multimodal Emotion Recognition

Fusion of Multiple Modalities

Approaches : Combines information from multiple sources, such as facial expressions, speech, physiological signals, and textual cues.[3]

Techniques:

Early Fusion: Combines features from different modalities before classification.

Late Fusion: Combines decisions from separate classifiers for each modality.

Hybrid Fusion: Integrates features at multiple levels of the model.

Key Models:

Deep Multimodal Neural Networks: Utilize separate networks for each modality, followed by a fusion layer for joint representation learning.

Attention Mechanisms: Focus on relevant parts of the data from each modality, improving performance in challenging scenarios.

Challenges: Requires large amounts of labeled multimodal data and increases computational complexity.

Pre-trained Models and Transfer Learning

Pre-trained Models: VGGFace, ResNet[7], and other models pre-trained on large face datasets (e.g., FaceNet) are fine-tuned on emotion recognition datasets.

Advantages: Significantly reduce training time and data requirements, while maintaining high accuracy.

Applications: Widely used in real-time systems for facial emotion detection, especially in applications like customer service, surveillance, and entertainment.

Challenges: Adaptability to specific datasets and generalization to diverse real-world conditions remain issues.

Current Trends and Future Directions

Lightweight Models for Real-time Applications

Techniques: Model compression methods like pruning, quantization, and distillation are used to create lightweight models for deployment on edge devices.

Example Architectures: MobileNet, EfficientNet, and TinyML models optimized for low-power devices.

Applications: Real-time emotion recognition in mobile devices, robotics, and embedded systems.

Challenges: Ensuring consistent performance across different environments, lighting conditions, poses, and occlusions.

Solutions:

Data Augmentation: Techniques like GANs (Generative Adversarial Networks) to generate diverse training samples.

Domain Adaptation: Adapting models trained on one domain (e.g., lab conditions) to work in another (e.g., real-world scenarios).

Ongoing Research: Developing models resilient to adversarial attacks and privacy-preserving techniques.

Ethical and Privacy Concerns

Issues: Emotion recognition technology raises concerns about privacy, consent, and potential misuse in surveillance or profiling.

Solutions: Implementing ethical guidelines, transparency in data usage, and developing algorithms that anonymize sensitive information.

5. PROBLEM STATEMENT

Human emotions play a crucial role in how we interact with the world around us. Our facial expressions often give away what we're feeling, even without us saying a word. In fact, two-thirds of our communication is non-verbal, with facial expressions being a key part of this. Emotions like joy, anger, and sadness can shape our responses and interactions, which is why understanding them is so important. However, translating this emotional understanding into technology is no simple task.

Emotion recognition systems aim to bridge this gap by using machine learning, sensors, and biometric analysis to interpret human emotions from cues like facial expressions, voice tones, and even physiological signals[1]. These systems have found their way into fields like education, healthcare, and marketing, where understanding how someone feels can lead to more personalized and effective interactions. Despite these advancements, current systems still struggle with several challenges.

One major hurdle is accuracy—emotion recognition systems perform differently across various datasets, and their results can be inconsistent. Another challenge is viewpoint variation, where the angle of a person’s face can make it harder for machines to accurately identify emotions.[4] Additionally, in real-time scenarios, things like wearing glasses can interfere with the system's ability to detect eyes, which are critical for emotion recognition.

We need systems that are not only better at recognizing emotions but can also do so in real-time, under various conditions, and with greater reliability. The demand for machines that can understand emotions is growing, and improving these systems will allow for more natural, human-like interactions between people and technology.

The challenge is clear: how can we make emotion detection systems more accurate, efficient, and capable of interpreting emotions in the real world? Overcoming these issues will unlock the potential for smarter, more empathetic technologies that enhance the way we interact with machines, and by extension, each other.

6. OBJECTIVE

The primary objective of this paper is to make a survey on ML and deep learning algorithms and other emotion detection techniques to develop a model that is capable of overcoming some limitations that are faced by traditional models such as the standard CNN , RVM and SVM .

One of the more common issues faced by such models is a stable accuracy across different datasets .Some other factors we need to focus on to make a high accuracy model for mixed datasets are

Generalization: Ensure the model performs well on diverse datasets, capturing a wide range of emotional expressions and contexts.

Robustness: Develop algorithms that can handle noise and variations in data, such as different languages, cultures, and modalities (text, speech, images).

Scalability: Design the system to efficiently process large volumes of data, making it suitable for real-time applications.

Ethical Considerations: Address privacy concerns and biases to ensure the system is fair and respects user confidentiality.

These objectives help in building a reliable and effective emotion detection system that can be applied in various real-world scenarios.

7. METHODOLOGY

7.1 Datasets

FER2013 database. The data collection used for the application was the FER2013 dataset from the Kaggle challenge on FER201338. The database is used to incorporate the Facial Expression detection framework. The dataset consists of 35,887 pictures, split into 3589 experiments and 28,709 images of trains. The dataset includes another 3589 private test images for the fnal test. Figure 1 shows the expression distribution of the FER2013 dataset.

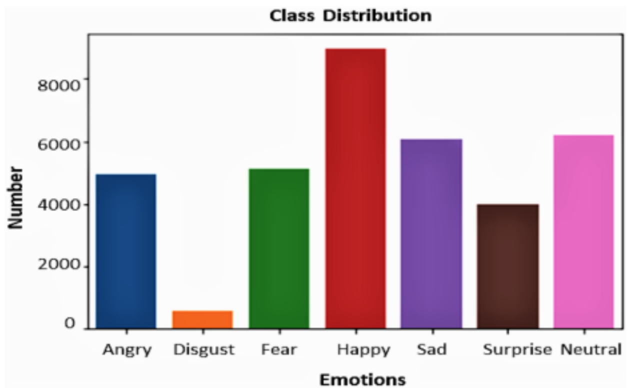

Fig 1

The extended Cohn–Kanade (CK+) database. Te CK+ database comprises 593 sequences from 123 people aged 18 to 30. Anger (45), neutral (18), disgust (59), fear (25), happiness (69), sadness (28), and surprise (83) are used to classify 327 sequences41. We selected 309 sequences from our trials that were classifed as one of the six fundamental facial expressions, eliminating contempt.

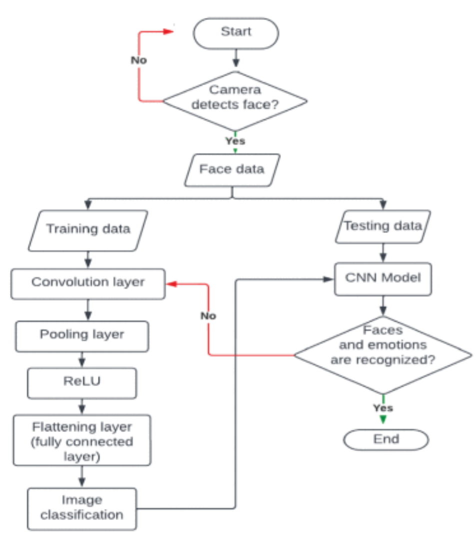

Fig 2

System flowchart of the proposed method (fusion feature extraction and fusion approaches with the proposed ConvNet Model) for emotion classification.

7.2Network Architecture:

The proposed system is based on the VGGNet architecture, chosen for its simplicity and strong performance in image-based tasks. The architecture consists of convolutional layers with 3x3 filters, max-pooling layers for down sampling, and fully connected layers for classification.

Depth: The depth and simplicity of VGG is popular. From sixteen to nineteen, the network comprises thirteen layers of convolution.

Small Filter Size: Three-by-three convolutions filters with a stride of 1 are the main filters used in VGG. Unlike larger filter sizes, this small filter size lets the network have a wider receptive field in fewer quantities.

Pooling Layers: VGG uses max-pooling layers after every two or three convolutional layers, each having a window size (i.e., k×k) of 2×2 and stride (s) equal to 2. This helps in reducing the feature map sampling but keeping the salient features.

Fully Connected Layers: In the VGG framework, in addition to the convolutional layers, there are three fully connected layers, which are mainly used for classification tasks. Rectified Linear Unit (ReLU) operation is applied on all layers except the output layer

Data Pre-processing Using OpenCV and Augmenter: Pre-processing the raw input image can involve augmenting, normalizing, and resizing to increase the diversity of training data thereby strengthening the model's robustness. An augmenter and OpenCV, a popular computer vision library are used for this purpose.[3]

Typically, normalize pixel values to a standard range of [0, 1] to ensure that the data input is uniform and does not lead to numerical instability in training. For our model we fused features from Fir13 and ck+ datasets and performed l2 regularization and normalization as shown in figure 2

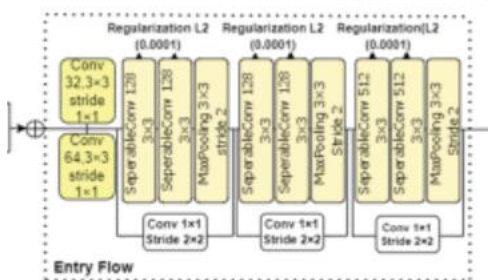

Fig 3

7.3 Model Training (FER Model): A Facial Emotion Recognition (FER) model is trained using the VGG architecture features [6] the data set errors associated with sensation detection. The CNN is trained using a training datasets, the ADAM optimization algorithm, and a special cross-entropy loss function. The goal is to minimize the loss function by varying the image weights several times, which will improve the image accuracy in facial expression recognition and segmentation.

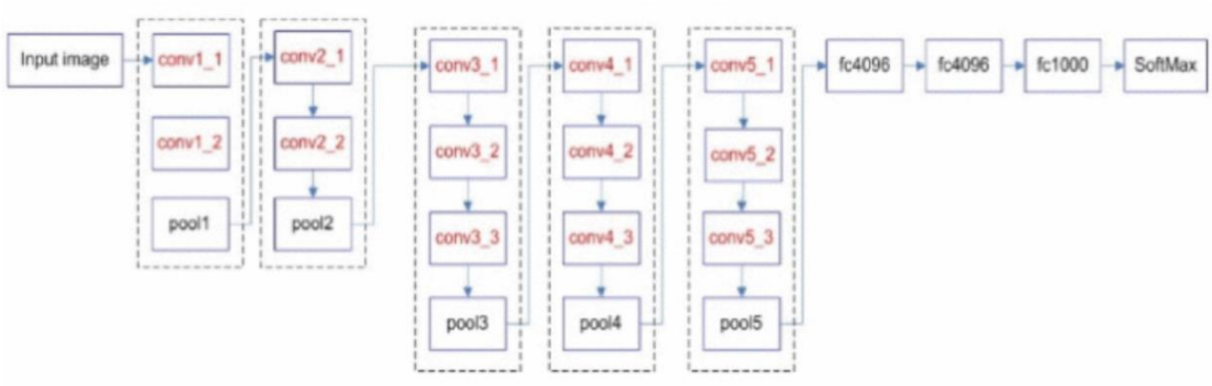

Fig 4

7.4 Fine-Tuning and Optimization: To even boost its performance further after the initial training, the model is optimized and fine-tuned. Some of this may entail increasing the number of epochs in training, using more complex optimization algorithms, or adjusting hyperparameters.

$$CE = -\sum_{i}^{C} t_i \log(f(s_i))$$

where C is the number of emotion labels, ti is the ground truth, and si is the model score for each class i in the fused data. The function F(si) refer to the activations applied to the output before the loss computation.

8.IMPLEMENTATION,RESULTS AND DISCUSSION

8.1 Training and testing setup

All mod All models were implemented in Pytorch with GPU 1xTesla K80, having 2496 CUDA cores and 12GB GDDR5 VRAM in Google Colaboratory. All datasets were first saved in Google drive and imported into Google Colaboratory.

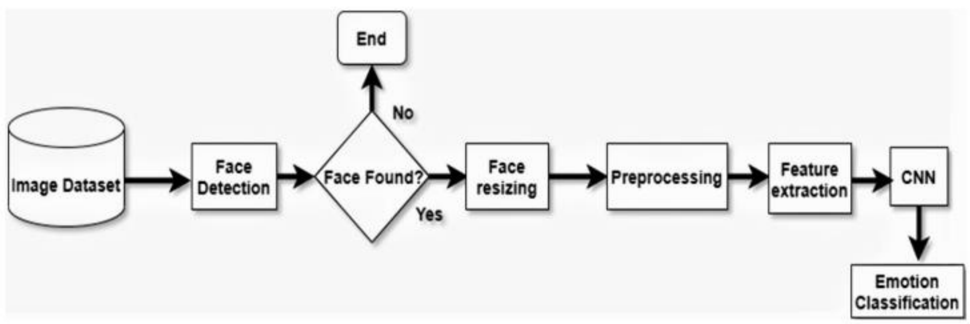

Fig 5

8.2 Recognition of Facial Expressions

Facial Expression Recognition (FER) mechanisms typically involve three primary steps: preprocessing, feature extraction, and emotion classification. These steps work in tandem to ensure accurate recognition of facial expressions. The overall system flow for emotion classification using the FER approach is illustrated in Figure .

1. Preprocessing

Preprocessing is a critical step in computer vision to prepare images for analysis by reducing variations and noise. Images in the dataset may contain inconsistencies such as lighting differences, scaling issues, and color variations. To address these challenges and improve the reliability and efficiency of the CNN algorithm, the following preprocessing techniques are applied:

Normalization: The image is normalized to eliminate inconsistencies and improve the clarity and accuracy of facial features.

Grayscale Conversion: Colored images are transformed into grayscale, as colored images can introduce complexity in algorithm processing. Grayscale simplifies the input to a single-channel image, allowing facial expression recognition to be performed independently of skin tone.

Resizing: Images are resized to remove unnecessary portions, reduce memory usage, and enhance computational efficiency.

2. Face Detection

Face detection is a foundational step in any Facial Expression Recognition system. This phase isolates the facial region from the input image. Viola and Jones' "Rapid Object Detection using a Boosted Cascade of Simple Features" (2001)[8] introduced Haar cascades as a powerful method for object detection. The Haar cascade classifiers identify specific facial regions (e.g., eyebrows, eyes, and forehead) while effectively removing unwanted background data. OpenCV's implementation of Haar cascade classifiers facilitates efficient face detection with high precision.

3. Feature Extraction

Feature extraction involves isolating critical features from the detected facial regions to identify expressions accurately. In this study, the Local Binary Pattern (LBP) method is employed for its simplicity, robustness, and widespread success in object recognition and facial expression detection tasks. The LBP method focuses on "uniform" patterns, which constitute the majority of patterns in the image dataset. These patterns enable efficient and reliable classification of facial expressions.

4. Emotion Classification

Once features are extracted, the image is classified into one of seven predefined emotion categories using a Convolutional Neural Network (CNN). The CNN model processes the extracted features to accurately predict the associated emotion.

Through the combination of preprocessing, face detection, and feature extraction, this FER system ensures robust and precise emotion classification, addressing key challenges such as background interference, lighting variations, and computational efficiency.

8.3 Experiments and results analysis Accuracy and complexity

Since the proposed model has been trained on a composite dataset, training accuracy above 95% and validation accuracy above 91% has been reached, which would be 95% after performing several epochs. It can be mentioned earlier that just after 30 epochs, the CNN model has a training accuracy of 95%, whereas CNN has taken further epochs to reach greater accuracy. A slight comparison of the suggested approach with other related works is seen in the Table 1. From the table, it can be ensured that the CNN approach is much better than adjusting any other technique or approach to the recognition of human emotions, and our proposed model demonstrates better work. The performance of the system established for facial emotion recognition from images was measured using three metrics: accuracy, specificity, and sensitivity.

Algorithm	**Accuracy (%)**	**Computational complexity**
Alexnet[6]	55–88	O4̂
VGG[7]	65–68	O9̂
GoogleNet[8]	82–88	O5̂
Resnet	72–74	O1̂6
FER (our proposed)	75–96	O4̂

Table 1

	Angry (%)	Disgust (%)	Fear (%)	Happy (%)	Sad (%)	Surprise (%)	Neutral (%)
Angry	**91.1**	0.2	0.2	0	4.2	3.8	2.5
Disgust	5.3	**89.7**	3.0	2.6	1.0	0	2.4
Fear	0.24	2	**88.6**	3.4	5.9	1.16	0.7
Happy	1.5	0	2.1	**95.9**	0	0	0.5
Sad	1.0	5.3	4.2	0	**88.9**	0	0.6
Surprise	0.24	3.0	0.63	0	0.53	**94.6**	1.0
Neutral	2.12	3.8	3.27	0.3	0.79	0.44	**92.3**

Table 2

The number of correctly classified data is divided by the total number of the data to calculate accuracy. True-positive (TP), true-negative (TN), false-positive (FP), and false-negative (FN) performance characteristics were used to compute the metrics as specified by following Eqs.

$$\text{Accuracy} = \frac{TN + TN}{TP + FP + FN + TN},$$

$$\text{Sensitivity} = \frac{TP}{TP + FN},$$

$$\text{Specificity} = \frac{TN}{TN + FP}.$$

Accuracy and loss graph

We recognize that the accuracy of training and validation are assessed to determine a model fitting. If there is a large difference between the two, the model is over-fitting. The accuracy of the validation should be equal to or marginally less than the accuracy of the preparation to be a better model. This work is also seen in Fig. 7, as the epoch improves the training accuracy is marginally higher than validation accuracy as we extended the layers and eventually introduced a few more convolution layers and several entirely related layers, rendering the network both larger and broader.

It seems like the lack of preparation can still be smaller than the loss of validation. In Fig. 6, We show the corresponding training versus validation failure. This means that the training loss reduces as the epoch grows, and the validation loss increases.

In addition, the validation data are always expected to decrease as the weights are adapted. Here, as the epoch grows in higher-order then we can expect a lower rate of validation loss than the training loss which we have already seen in the last stages of the figure. Therefore, this model is well suited to the training results.

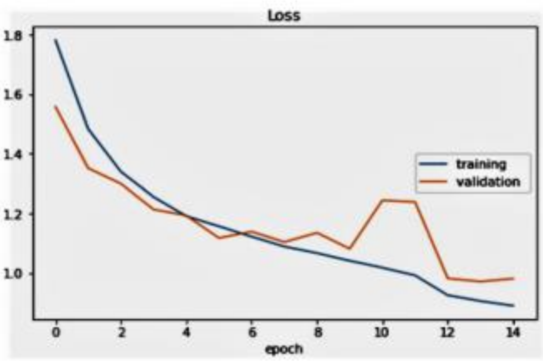

Fig 6

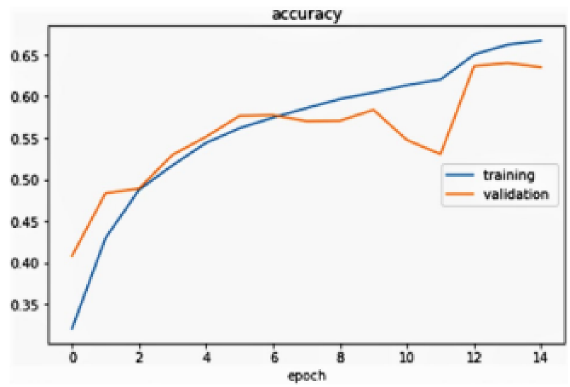

Fig 7

9. CONCLUSION AND FUTURE WORK

Fusion features were employed in the proposed study, and the input data were fused using CNN features. Based on the CK+ dataset, the proposed technique was found to have a better accuracy (98.13%) than other recent methods. In comparison to the current techniques, the other datasets equally obtained the best accuracy. The FER2013 also outperforms the previous studies in terms of classification accuracy.

	Angry (%)	Disgust (%)	Fear (%)	Happy (%)	Sad (%)	Surprise (%)	Neutral (%)
Angry	**98.61**	0	1.33	0	0.96	0	0
Disgust	0	**96.46**	0.27	0	1.97	2.30	0
Fear	0	0	**99.60**	0	0	0.40	0
Happy	0.31	1.95	0.32	**97.51**	0.30	0	1.61
Sad	1.11	1.29	0	1.10	**96.40**	0	1.03
Surprise	0.25	0	0	0.23	1.89	**97.28**	0.35
Neutral	1.12	1.80	1.27	1.97	2.17	0.42	**98.25**

Table 3

9.1 Future Work

We have found seven distinct and unique emotion classes (fear, happy, angry, disgust, surprise, sad and neutral) in this article for emotion classification. Tere is no overlap between groups as our model perfectly fts the data. The presented research has achieved a very good average classification accuracy of emotions over 92% and also achieved 96% for random division of data in a minimal number of epochs. Based on the results from existing research studies, our model has been reflecting better in the real-time. We, however, plan to work with complex type or mixed group of emotions, such as shocked with pleasure, surprised by frustration, dissatisfied by anger, surprised by sorrow, and so on. In this analysis, we conclude that due to certain variables, their proposed model is only on average. For the future, we will continue to focus on enhancing the consistency of each CNN model and layer and also try to examine new feature or method to be fused with CNN. The future study involves examining multiple forms of human variables such as personality characteristics, age, and gender that affect the efficiency of emotion detection.

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